

CS671 – Deep Learning and Its Applications

Technical Documentation Report

Model: Pix2Pix (Brain)

Project Group: 5 Region: Brain Modality: Paired (CT → MRI)

1. ARCHITECTURAL OVERVIEW

Pix2Pix is a conditional Generative Adversarial Network (cGAN) designed for supervised image-to-image translation. In the Brain CT-MRI task, the model learns a direct mapping from source to target modality by optimizing a combination of adversarial loss and a pixel-wise reconstruction loss, leveraging the spatial alignment of the paired dataset.

1.1 Generator Architecture (U-Net 256)

The generator uses a symmetric U-Net architecture with 8 levels of downsampling and upsampling, incorporating skip connections to bypass the bottleneck for high-frequency structural details.

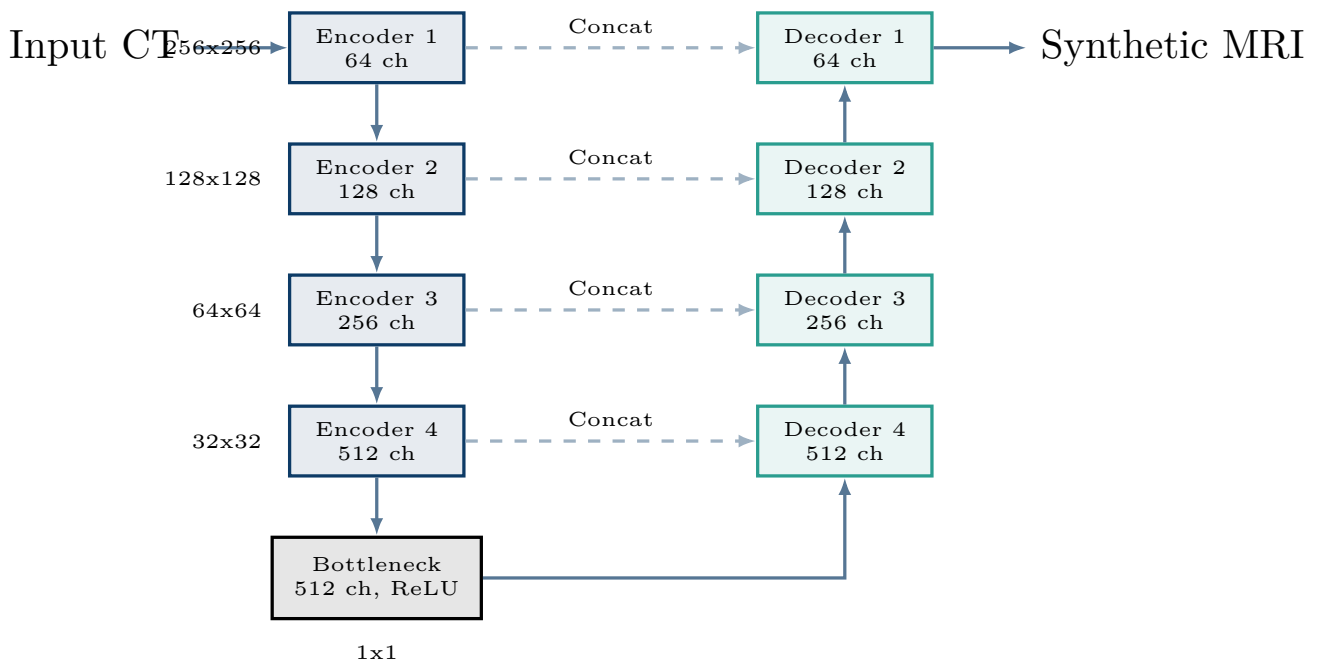


Figure 1: U-Net 256 Generator: Explicit concatenation skip connections for structural fidelity.

Encoder Stage: Each encoder block consists of a 4×4 convolution with stride 2, followed by Batch Normalization and a LeakyReLU activation. This progressively reduces the spatial resolution while extracting semantic features such as brain tissue density and skull geometry.

Skip Connections & Information Flow: Medical images, especially brain scans, require exact spatial correspondence. By concatenating the feature maps from the i -th encoder layer directly to the $(n - i)$ -th decoder layer, we provide the decoder with "shortcuts" to low-level anatomical information that would otherwise be compressed or lost in the bottleneck. This ensures that synthesized MRI images align perfectly with the original CT bone structure.

Decoder Stage: The decoder uses transpose convolutions to upsample the features. Each block (except the last) includes Batch Normalization and Dropout (0.5 during training) to prevent overfitting and encourage variety in the generated textures.

1.2 Discriminator Architecture (70x70 PatchGAN)

The discriminator is conditioned on both the input CT and the candidate MRI, forcing it to judge not just the "realness" of the MRI, but also its alignment with the CT.

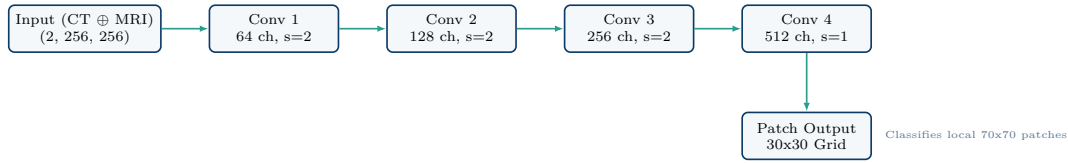


Figure 2: Conditional PatchGAN: Concatenating CT and MRI as input for alignment-aware discrimination.

2. TRAINING DYNAMICS

The objective function of Pix2Pix combines a conditional GAN loss with an L_1 reconstruction loss to enforce both perceptual realism and pixel-wise accuracy.

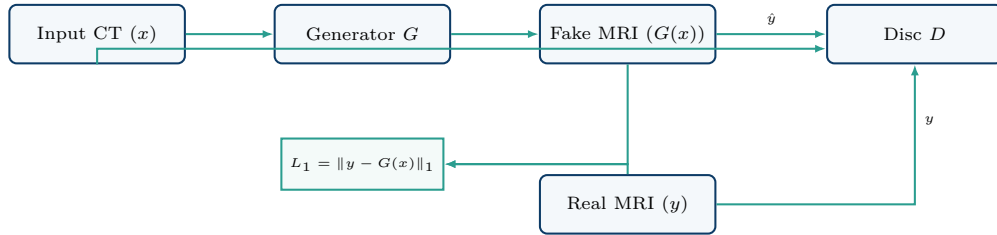


Figure 3: Training workflow: Concurrent optimization of GAN and L_1 losses.

Loss Formulation: The total loss is defined as: $G^* = \arg \min_G \max_D \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G)$.

- **L_1 Loss:** Encourages the generator to produce images that are pixel-for-pixel similar to the ground truth MRI. This is the primary driver for anatomical accuracy.
- **Adversarial Loss:** Encourages the generator to produce high-frequency details (textures) that the L_1 loss (which tends to blur) might ignore.

3. INFERENCE PIPELINE

Inference is performed as a deterministic forward pass through the trained U-Net generator.

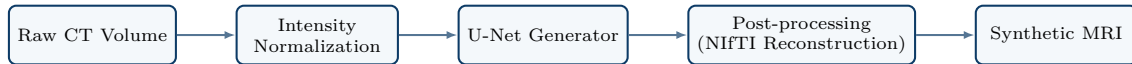


Figure 4: Inference flow: Direct translation of CT volumes to the MRI domain.

The inference system handles 3D NIfTI files by iterating through the axial stack. Each slice is normalized to the $[-1, 1]$ range used during training. The final output is rescaled to match the original MRI modality intensity distribution, facilitating direct comparison in clinical software.